SIM-02B.2-A Review Transmittal

Ulmo / Picket, PCK-SIM-001: adaptive noise estimation and false-alarm control (review gate before SIM-02B.2-B)

To	Author of the SIM-02B.1 review and SIM-02B.2 authorization
Subject	Verified SIM-02B.2-A adaptive noise-estimation and false-alarm results for review
Date	2026-11-09
Repository	`keithatulmo/picket-sim`, branch `claude/beautiful-darwin-vbeoue`
Run of record	`20261109T003718Z_27abcd69_73fbe35` (code commit `73fbe35`, clean tree)
Prepared by	Claude Code (AI-generated engineering work product). Requires review by a qualified acoustic engineer before any of it informs requirements.
Requested action	Review the SIM-02B.2-A results; decide the items in Section 7 before SIM-02B.2-B begins

1. Disposition of the SIM-02B.1 review

The SIM-02A closeout and SIM-02B.1 acceptance are recorded in ADR-0018 together with D01 to D07, the A/B/C staging and the additional requirements. The SIM-02B.1 run of record, errata, snapshots and package are unchanged; this cycle added `results/sim02b2a/ERRATA.md` only. SIM-02B.2-B and -C have not started.

2. SIM-02B.2-A deliverables (review Section 6, per stage)

#	Required output	Evidence
1	Plan and governing equations	`docs/SIM-02B2A_plan.md` (equations, design, priorities, measured compute budget); ADR-0019 (normalizers, partitions, ground-truth isolation), ADR-0020 (calibration methods), written before the run
2	Source and tests	`src/picket_sim/noise_estimation.py`, `qforms.py`, `experiments/sim02b2a.py`; 35 tests in `tests/test_sim02b2a_*.py`
3	Provenance and limitations	`tables/parameters_by_provenance.csv`; `docs/assumptions.md` AS-30 to AS-36; report Section 5
4	Versioned configurations	`configs/sim02b/sim02b2a_baseline.yaml`, generated by `tools/make_sim02b2a_config.py` (a test requires the committed file to equal the generator output); resolved configuration in `record.json`
5	Numerical verification and independent comparisons	Exact CFAR laws against the measured A0 loss (within 0.13 dB); tail methods against empirical thresholds (33 of 36); ORACLE results against SIM-02B.1 (343 of 348 consistent); estimator laws and contamination ceilings (`reports/SIM-02B2A_supplement.md`); partial rerun (`make sim02b2a-repro-check`)
6	Comparative results with uncertainty	`reports/SIM-02B2A_summary_tables.md`; run tables `realized_pfa.csv`, `snr_required.csv`, `detection_loss_vs_oracle.csv`, `gain_vs_a0.csv`, `estimation_error.csv`, `pd_curves.csv.gz`
7	Defect and errata register	`results/sim02b2a/ERRATA.md`; Section 4 below
8	Engineering assessment	Report Sections 0 and 6

3. Additional requirements of the authorization

Requirement	How it is met
A0 in all comparisons	A0 runs in every scenario and cell; gains use the same normalizer on A0 and the array
Pd 0.5 at Pfa 1e-3 for continuity, plus Pd 0.9 and other Pfa	Every SNR-required row at Pd 0.5 and 0.9 and Pfa 1e-2, 1e-3, 1e-4 where a threshold exists
Per-window Pfa against false alerts per hour	Every realized-Pfa row carries false alerts per hour; `tables/false_alerts_per_hour.csv`; report finding 6
No operational conclusions from stationary Gaussian noise	Report front matter, findings and Section 5
No contamination of background estimates by target-bearing evaluation data	Reference windows are generated inside each trial with separate seeds; the target enters reference windows only in the CONTAM scenario, which measures that effect
False-alarm, threshold and detection uncertainty reported separately	Report Section 2; separate columns in every table
True covariance only as an oracle upper bound; detectors without ground truth	ORACLE labeled as a bound; adaptive normalizers take data arrays only (ground-truth isolation test)
Independent training, calibration and evaluation partitions with guards	ADR-0019; guard of 2 windows (8 in CONTAM) and 2 bins

4. Defects found and corrected during this cycle

- **Run table status column (recorded in `results/sim02b2a/ERRATA.md`).** The `pfa_status` column of the run tables used a Wilson interval of the evaluation set only. That is the same defect class recorded for the SIM-02B.1 self-check, repeated in a new column: it ignores threshold sampling error and flags 3 exceedances in 10,000 at Pfa 1e-4 as liberal, which happens with probability 0.08 under correct calibration. No result value is affected; the run's self-check and the report use the exact predictive laws. Code after the run adds exact tails and an exact status column, with a test.
- **Self-check count and figure labels (ERRATA).** The method-agreement self-check counted identical NOMINAL rows twice (66 of 72; 33 of 36 distinct statistics); figure 6 lacked axis labels. Corrected in code after the run.
- **Caught by the claims checker before delivery.** The draft report said the not-estimated thresholds were all adaptive (108 of the 360 are ORACLE and NOMINAL rows for statistics other than the tonal maximum), quoted the coherence-mismatch maximum as 2.40 times target (it is 2.93), and miscounted the increment tests. All corrected.
- **SIM-02B.1 claims checker.** Its check of "214 tests at the packaging baseline" counted the tests of whatever tree it ran in, so it could pass only at that baseline; the SIM-02B.2-A package build caught this when it ran `make sim02b1-claims` from the new snapshot. It now checks the inventory recorded in the SIM-02B.1 package (committed as a test fixture) and requires every test in it to still exist. No SIM-02B.1 number changes.
- **Regression tool.** Its first version compared SIM-02B.1 union-bound rows as estimates and reported 21 inconsistencies; bound rows are now tested one-sidedly (all 48 hold).
- **Before the run of record** (found in self-review and the smoke run, none in a delivered result): the contamination normalizer had no threshold (it now shares the threshold of the same estimator); a block approximation in the conditional Monte Carlo was replaced by every-bin estimation with common random numbers and its interval sign corrected; a reference-window axis error and two configuration-path errors were fixed; ADR-0020 was corrected to name the implemented matrix-exponential method.

5. Reviewer decisions D01 to D07: disposition

Decision	Disposition
D01 adaptive noise estimation	Partly done. Time-reference and frequency-reference estimators (cell averaging and order statistic) against ORACLE and NOMINAL, with estimation loss, contamination, nonstationarity, mismatch, false-alarm stability and processing cost. **Not done: robust multichannel covariance estimation.** Beam-domain cell-averaging references equal the quadratic form of the sample covariance with conventional weights, but no robust covariance estimator, whitening or covariance-adaptive beamformer was tested (decision 2)
D02 coherence collection	`docs/coherence_collection_plan.md` with computed design tables; no collection performed; published data stay interim references
D03 Wenz nominal	Retained; SHAPE (shipping and wind variants) and STEP (level) scenarios
D04 synchronization ranges	SIM-02B.2-C, as authorized
D05 AVS	No purchase; evidence file kept; loaned-sensor outline with objectives and cost structure (`docs/avs_feasibility.md` Section 7; prices need quotes)
D06 trial budget	Measured budget and priorities first (plan Sections 6 and 7); exact laws, the exact quadratic-form law, conditional Monte Carlo and sequential calibration with a stopping rule; bounds labeled; the 139 SIM-02B.1 cases were not scaled
D07 beam grid	SIM-02B.2-B; the 24-beam grid kept for regression

6. Principal findings

Stationary circular Gaussian noise; Wenz-type spectra with assumed parameters; assumed coherence and scenario parameters (4 published records, 0 measured, 27 assumed). Operating point: 200 Hz to 2 kHz, 1 s windows, Pfa 1e-3 per window, Pd 0.5 (report Section 0).

1. False-alarm control with estimated backgrounds holds under the nominal design: 1,170 rows, 4.3 % of p-values below 0.05, none below 0.001.
2. Estimation costs tonal detection 2.10 dB on A0 and 2.51 to 2.63 dB with beam search, broadband 0.11 to 0.47 dB; the A0 loss matches the exact CFAR law within 0.13 dB.
3. The array advantage over A0 survives estimation: P8-C beam search keeps 2.83 to 3.05 dB of its 3.42 dB tonal gain. Below 200 Hz no compact candidate beats A0 (-0.26 to 0.72 dB).
4. Time references fail under a 1 dB level step (tonal false alarms 3.50 to 11.25 times target), under interference in the reference windows (4.52 to 5.42 dB extra loss) and against persistent targets (Pd never reaches 0.5); frequency references hold in all three but cannot serve broadband statistics.
5. No level estimator repairs a coherence mismatch: under flow-like coherence, 121 of 132 incoherent rows give no false alarm at Pfa 1e-2 and beam search runs up to 2.93 times target.
6. Pfa 1e-3 on 1 s windows is 3.6 false alerts per hour per detector.

Nothing here selects an architecture, estimates a detection range or sets a hardware requirement.

7. Decisions requested before SIM-02B.2-B

1. **Exit requirement.** Accept that SIM-02B.2-A meets its exit requirement within the stated model scope, and authorize SIM-02B.2-B.
2. **Robust multichannel covariance estimation (D01 gap).** Complete it as a SIM-02B.2-A supplement before SIM-02B.2-B (sample covariance with shrinkage and outlier-robust estimates, whitened and covariance-adaptive statistics, same scenarios and threshold methods), or fold it into SIM-02B.2-B.

3. **Normalizers for SIM-02B.2-B.** Carry frequency references for tonal, harmonic and chirp statistics and time references with the longer guard for broadband statistics, plus ORACLE as a bound and A0 with the same normalizer; or name a different set.
4. **Multichannel threshold law under coherence uncertainty** (finding 5): wait for D02 data, add a data-driven threshold calibration to SIM-02B.2-B, or both.
5. **Array thresholds at Pfa 1e-4.** Fund empirical calibration (at the n Pfa = 10 minimum, 2.5 to 3.0 core-hours per array in the 200 Hz to 2 kHz cell, from this run's timings; more for a 0.25 dB interval), or accept Pfa 1e-3 as the array reporting floor until a tail method covers those statistics.
6. **Target-aware reference censoring** (excluding windows that already hold a detection) as a SIM-02B.2-B item, given the contamination ceiling.
7. **D02:** approve the collection plan in principle and name candidate sites and host access.
8. **D05:** approve requesting loan and facility quotes for the bench evaluation outline.

8. Suggested review path (about 60 minutes)

1. This transmittal, Sections 4, 6 and 7.
2. Report Section 0 (findings), Figures 1 to 4, Sections 5 and 6.
3. `reports/SIM-02B2A_summary_tables.md` S2, S3a and S6 (calibration, robustness, contamination).
4. `results/sim02b2a/ERRATA.md` and `05_verification/` (tests at both commits, snapshot checks).
5. `docs/coherence_collection_plan.md` (D02) and `docs/avs_feasibility.md` Section 7 (D05).
6. Optional: `make sim02b2a-claims` and `make sim02b2a-repro-check` from the source snapshot.